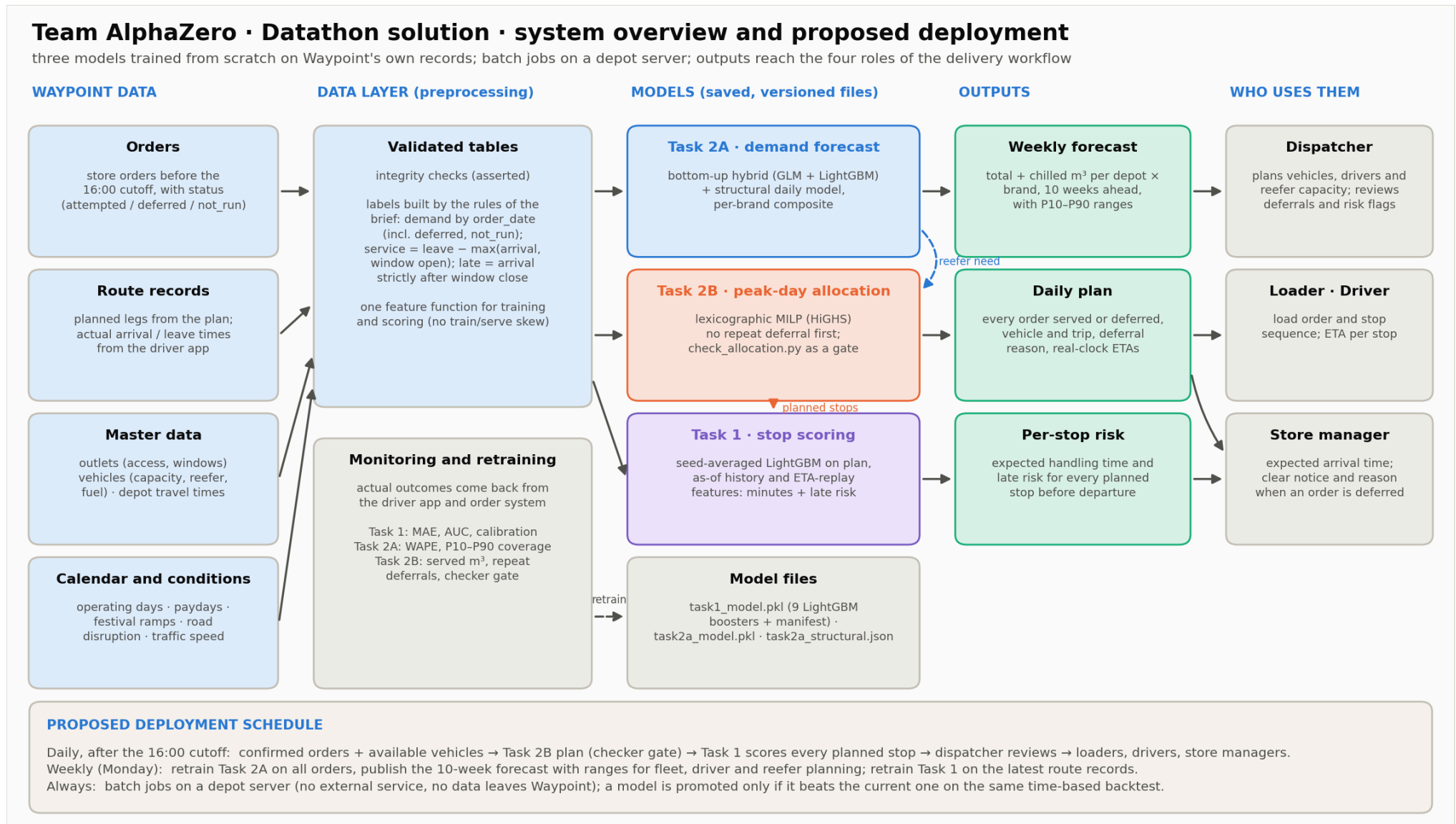


Architecture diagrams

Tech-Triathlon 2026 Datathon · Team AlphaZero · models, preprocessing pipeline and proposed deployment

1. System overview and proposed deployment



One data layer (validated tables, labels built by the brief's rules, one feature function per task) feeds three models that run as batch jobs on a depot server, with no external service and no data leaving Waypoint. **Daily** after the 16:00 cutoff: Task 2B plan → official checker gate → Task 1 scores every planned stop → the dispatcher reviews. **Weekly**: Task 2A and Task 1 are retrained, monitored, and replace the current models only if they win the same time-based backtest.

2. Task 1 · service time and lateness

Team AlphaZero · Datathon Task 1 · architecture

service time and lateness per planned delivery: LightGBM trained from scratch on the supplied data; no pre-trained models, no APIs

1 · SUPPLIED DATA

Training records

deliveries_train.csv
route_legs_train.csv
(with actual times)
91,894 dispatched orders

Test inputs

task1_test_inputs.csv
route_legs_test.csv
(planned times only)
5,014 deliveries

Reference tables

outlets · vehicles
service_allowance · calendar
road_conditions
traffic_speed

2 · JOIN, CHECKS, LABELS

Join and checks

order ↔ route leg on
route_id + seq_in_route
(one-to-one: each order is its
own stop); times → minutes

87 integrity and label checks,
all asserted

Labels (training only)

service_min = leave - max(arrival,
window open) (early vehicles wait)

is_late = arrival > window close
(strictly; 279 arrive exactly
at close and are on time)

actual times never become
features; they only feed the
past-history aggregates

3 · FEATURES (96 lateness / 91 service)

Plan and context (47 + 8 categorical)

order size · dock · access · allowance
window · stop position · planned slack
route load · traffic at the leg's hour
road disruption · calendar

History h_* (31), as of a cutoff

outlet, vehicle, district, depot history
from labelled rows strictly before the
cutoff (block start; test: 16 Feb 2026);
size-scaled minutes per unit and kg

ETA replay η_* (10)

each planned route replayed stop by
stop: expected travel, window waits,
expected service (size-aware) →
expected arrival, delay and slack

4 · MODELS AND VALIDATION

Service-time model

$0.75 \times \text{LightGBM on } \log(1 + \text{minutes})$
+ $0.25 \times \text{LightGBM squared error}$
3 seeds each · MAE 3.89 min
(planning allowance 7.32)

Lateness model

LightGBM binary, 3 seeds averaged
log loss 0.142 · AUC 0.979
calibrated (ECE 0.006),
no post-hoc calibration needed

Time-based validation

3 six-week folds + 2 folds no decision
used; early stopping on the last 6
weeks before each fold, refit $\times 1.1$;
baselines · ablations · search · gates

5 · OUTPUTS

Saved models

models/task1_model.pkl:
9 LightGBM boosters
(3 models $\times$ 3 seeds) + manifest
(also as text: models/task1/)

submission_task1.csv

5,014 rows, template order
service > 0 · prob in (0, 1)

Final notebook cell

reloads the models,
rebuilds test features from
the raw files, prints inputs
+ predictions

PROPOSED DEPLOYMENT

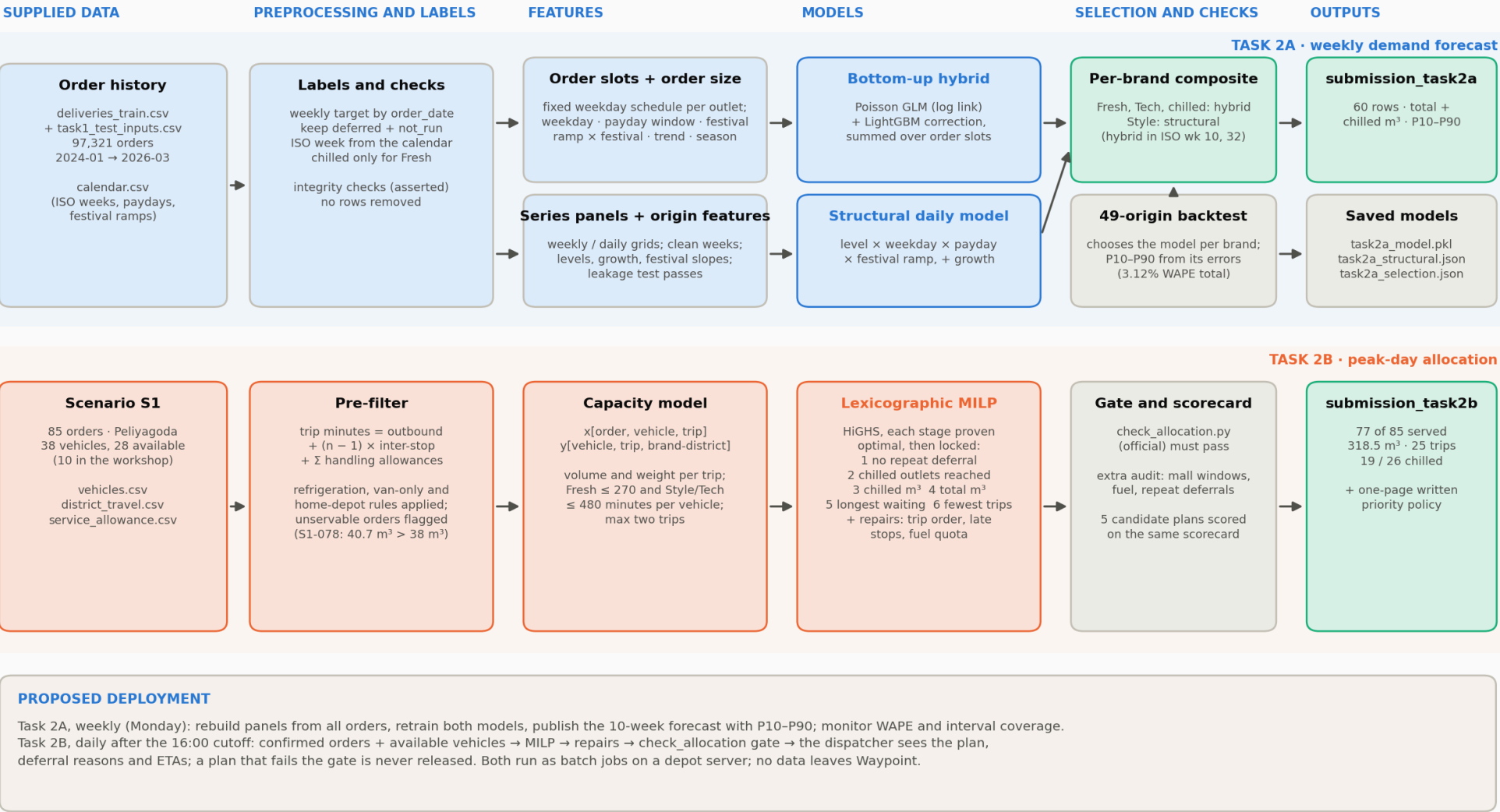
Daily, after the plan is built (16:00 cutoff): history is refreshed as of the run date, every planned route is replayed, and each stop gets its handling time and late probability; high-risk stops are flagged to the dispatcher before departure, ETAs go to drivers and store managers. Weekly: retrain on the latest route records; promote the new models only if they win the same time-based backtest; monitor MAE and calibration.

Orders join their route legs one-to-one, and the two labels are built from the actual times (training only). Features are known before departure: the plan, as-of history from labelled rows strictly before a cutoff (block start in training, 16 Feb 2026 for the test), and an ETA replay of every planned route. Seed-averaged LightGBM models give `pred_service_min` (0.75 log-target + 0.25 squared-error blend) and `pred_late_prob`. On six-week time folds: service MAE 3.90 min against 7.39 for the planning allowance; lateness log loss 0.143, AUC 0.979, calibrated; two folds that no decision used confirm it (3.88 min, 0.140).

3. Tasks 2A and 2B

Team AlphaZero · Datathon Tasks 2A and 2B · architecture

models, preprocessing pipeline and proposed deployment (all models trained from scratch on the supplied data; no pre-trained models, no APIs)



Task 2A. The weekly target is built by `order_date` from every order. Two model families run in parallel: a bottom-up hybrid (Poisson GLM + LightGBM) on the order slots, and a structural daily model on leakage-tested series panels. A 49-origin rolling backtest picks the family per brand and sets the P10-P90 ranges: 3.12% WAPE on total volume, 2.07% on chilled. **Task 2B.** A pre-filter applies the access and refrigeration rules. An exact lexicographic MILP applies the priority policy, operational repairs fix trip order, late stops and fuel, and the official checker is a hard gate before any plan is released: 77 of 85 orders served.